Example 7.40

Evaluate : $\lim_{x\to 0^+} x\log x$.

Solution

This is an indeterminate of the form $(0\times\infty)$. To evaluate this limit, we first simplify and bring it to the form $\left(\frac{\infty}{\infty}\right)$ and apply l'Hôpital Rule. Thus, we get

$$\lim_{x\to 0^+} x\log x = \lim_{x\to 0^+}\left(\frac{\log x}{\frac{1}{x}}\right) \quad \left(\frac{\infty}{\infty} \text{ form}\right)$$

$$= \lim_{x\to 0^+}\left(\frac{\frac{1}{x}}{-\frac{1}{x^2}}\right) = \lim_{x\to 0^+}(-x) = 0 .$$

Example 7.41

Evaluate : $\lim_{x\to\infty}\left(\frac{x^2+17x+29}{x^4}\right)$.

Solution

This is an indeterminate of the form $\left(\frac{\infty}{\infty}\right)$. To evaluate this limit, we apply l'Hôpital Rule.

Then, we have $\lim_{x\to\infty}\left(\frac{x^2+17x+29}{x^4}\right) = \lim_{x\to\infty}\left(\frac{2x+17}{4x^3}\right)$

$$= \lim_{x\to\infty}\left(\frac{2}{12x^2}\right) = 0 .$$

Example 7.42

Evaluate : $\lim_{x\to\infty}\left(\frac{e^x}{x^m}\right), m\in N$.

Solution

This is an indeterminate of the form $\left(\frac{\infty}{\infty}\right)$.

To evaluate this limit, we apply l'Hôpital Rule m times.

Thus, we have $\lim_{x\to\infty}\frac{e^x}{x^m} = \lim_{x\to\infty}\frac{e^x}{m!}$

$$= \infty .$$

7.5.4 Indeterminate forms $0^0, 1^\infty$ and ∞^0

In order to evaluate the indeterminate forms like this, we shall first state the theorem on the limit of a composite function.

Theorem 7.6

Let $\lim_{x\to\alpha} g(x)$ exist and let it be L and let $f(x)$ be a continuous function at $x = L$. Then,

$$\lim_{x\to\alpha} f(g(x)) = f\left(\lim_{x\to\alpha} g(x)\right).$$

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